

SHAHAD AL-MATRAFI

ANALYTICS ENGINEER — DATA PIPELINES, MODELING, ML, AGENTIC AI

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Analytics Engineer with hands-on experience building end-to-end data pipelines, multi-layer data models (Kimball methodology), and centralized BI reporting for executives. Comfortable working across the Data manipulation tools, from Python/SQL pipeline automation to DAX/Power Query semantic models to production ML pipelines (91% F1 fraud detection model) and a recommendation engine built from scratch on association rule mining. Looking to bring this pipeline-to-insight ownership into a role spanning centralized reporting and ML-aware data engineering for recommendation systems.

CORE EXPERTISE

- Programming & Query:** Python (Pandas, automation), SQL, Power Query M, DAX, R
- Data Modeling:** Multi-layer modeling, Star Schema, Kimball Methodology, relational & fact table design
- BI & Reporting:** Power BI (advanced), Tableau, executive-level centralized reporting & metric definition
- Pipelines & Automation:** Python/Pandas data-cleaning pipelines, Power Automate, API integrations, scheduled refresh automation
- ML & Recommendation Systems:** Scikit-learn, XGBoost, feature engineering, association rule mining, collaborative filtering, Agentic AI / LLM integration
- Tools & Systems:** Oracle ERP, Git/GitHub, Flask, React, Firebase

PROFESSIONAL EXPERIENCE

Data Analyst Oct 2024 – Present
Saudi Services Company Limited (SSCL) · Makkah, Saudi Arabia

- Own end-to-end analytics for corporate fleet and asset data: designed a multi-source data model with conditional logic and optimized DAX to govern and reconcile vehicle records across disparate registries.
- Built Python/Pandas pipelines to clean and standardize raw data feeds, replacing manual spreadsheet processing and removing recurring data-entry errors.
- Reduced Power BI dashboard load time by **29%** by restructuring the underlying data model and rewriting DAX, improving responsiveness across all executive-facing reports.
- Designed and deployed **10+ production dashboards** across logistics, HR, and operations, replacing ad hoc Excel reporting as the primary reporting layer for department heads.
- Engineered a consolidated "Consumption" budget pipeline pulling inputs from **60+ departments** into a single automated report, eliminating a manual annual forecasting process.
- Supervised 5 on-the-job trainees (split into two cohorts) in data analytics, managing task allocation and code reviews for pipeline development.

KEY TECHNICAL PROJECTS

Co-Thinker — Agentic AI Tutoring Workspace Jun 2026
React · Gemini 1.5 Pro API · Agentic Workflows

- Architected a spatial AI tutoring agent using React and the Gemini 1.5 Pro API, with a custom logic layer linking on-screen annotations to document coordinates for context-aware responses.
- Designed an agentic workflow with a Socratic tutoring pattern — the agent withholds direct answers and instead generates targeted hints, guiding the student's own reasoning.

Fraud Detection — Python ML Pipeline + Power BI Aug 2025
Python · Scikit-learn · XGBoost · Power BI

- Built a full fraud-detection pipeline handling imbalanced transaction data, benchmarking 4 algorithms, and achieving an **F1 score of 91%** on the best-performing model.
- Surfaced model predictions and risk scores in a live Power BI dashboard.

Jayil — Tourism Recommendation Engine Graduation Project, 2023
Python · R · Flask · Association Rule Mining

- Implemented a customized Disjunctive Association Rule Mining algorithm from academic research, coded from scratch without high-level libraries, to model non-traditional item-to-item associations.
- Re-engineered core data structures to reduce time complexity and accelerate recommendation latency for single- and group-travel itinerary generation.
- Built the full pipeline end-to-end: Google Maps data extraction (R) → recommendation algorithm (Python) → Flask API → Flutter app with Firebase backend and Mapbox maps.

CERTIFICATIONS & TRAINING

- Microsoft Certified: Azure AI Engineer Associate** — In Progress (95%+ complete)
- DAX in Semantic Models & Model Data with Power BI** — Microsoft Learn, Aug 2025
- Google Data Analytics** — Coursera, Sep 2024 | **Machine Learning: AI, Python & R** — Udemy, Aug 2024

EDUCATION

Bachelor of Computer Science
Umm Al-Qura University · Sep 2019 – Nov 2023
First Class Honours · GPA 3.81/4.00